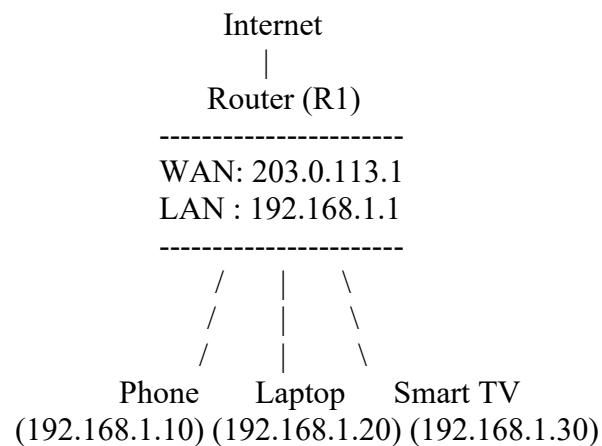


N+ - Network Fundamentals and building network

❖ IP routing and routing protocols

[Q-1] Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.

[ANS]



➤ Device IP Addresses

Device	IP Address
Router (LAN Interface)	192.168.1.1
Router (WAN Interface)	203.0.113.1
Phone	192.168.1.10
Laptop	192.168.1.20
Smart TV	192.168.1.30

[Q-2] Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made. Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.

[ANS]

➤ Network Topology

Laptop Router R1 Router R2 Friend's PC
 192.168.1.10 -----> 192.168.1.1/10.0.0.1 -----> 10.0.0.2/192.168.2.1 -----> 192.168.2.20

Device	IP Address
Laptop	192.168.1.10
R1 LAN	192.168.1.1
R1-R2 Link	10.0.0.1
R2-R1 Link	10.0.0.2
R2 LAN	192.168.2.1
Friend's PC	192.168.2.20

- **Step 1:** Laptop wants to send data to Friend's PC (192.168.2.20).
- **Step 2:** Laptop checks destination IP.
 - Destination = 192.168.2.20
 - Laptop subnet = 192.168.1.0/24
 - Since destination is outside local subnet, packet is sent to default gateway **192.168.1.1 (R1)**.
- **Step 3:** Router R1 receives the packet.
 - R1 checks its routing table.
 - R1 finds route to 192.168.2.0/24 via Router R2 (10.0.0.2).
 - Packet forwarded to R2.
- **Step 4:** Router R2 receives packet.
 - R2 sees destination network 192.168.2.0/24 is directly connected.
 - Packet forwarded directly to Friend's PC.
- **Step 5:** Friend's PC receives packet successfully.

[Q-3] Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

[ANS]

➤ Network Layout

Branch	City	Network
Branch A	Ahmedabad	192.168.10.0/24
Branch B	Mumbai	192.168.20.0/24
Branch C	Delhi	192.168.30.0/24

➤ Central Router Routing Table

Network Address	Subnet Mask	Next Hop	Interface
192.168.10.0	255.255.255.0	Direct	G0/0
192.168.20.0	255.255.255.0	10.0.1.2	S0/0/0
192.168.30.0	255.255.255.0	10.0.2.2	S0/0/1

➤ Example

- Ahmedabad orders are handled locally.
- Mumbai traffic forwarded through next hop 10.0.1.2.
- Delhi traffic forwarded through next hop 10.0.2.2.

[Q-4] Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

[ANS]

Feature	Static Routing	Dynamic Routing
Configuration	Manual	Automatic
Scalability	Small networks	Large networks
Updates	Manual	Automatic

➤ Static Routing

- **Advantages**

1. Simple to configure in small networks.
2. No routing protocol overhead.
3. More secure because routes are manually controlled.

- **Disadvantages**

1. Difficult to manage in large networks.
2. Network changes require manual updates.
3. No automatic failover.

➤ Dynamic Routing

- **Advantages**

1. Automatically discovers routes.
2. Adapts to network changes.
3. Suitable for large organizations.

- **Disadvantages**

1. More CPU and memory usage.
2. More complex configuration.
3. Routing updates consume bandwidth.

[Q-5] Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does. Hint: Paste the AI-generated config and annotate each part with a short description.

[ANS]

➤ Router R1 Configuration

- Router(config)# router ospf 1
- Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
- Router(config-router)# network 10.0.0.0 0.0.0.3 area 0

➤ Router R2 Configuration

- Router(config)# router ospf 1
- Router(config-router)# network 192.168.2.0 0.0.0.255 area 0
- Router(config-router)# network 10.0.0.0 0.0.0.3 area 0

Command	Description
router ospf 1	Starts OSPF process ID 1 on the router.
network 192.168.1.0 0.0.0.255 area 0	Advertises the 192.168.1.0/24 network into OSPF Area 0.
network 10.0.0.0 0.0.0.3 area 0	Advertises the link between routers into OSPF Area 0.
network 192.168.2.0 0.0.0.255 area 0	Advertises the 192.168.2.0/24 network into OSPF.
area 0	Specifies the OSPF backbone area used for routing.

➤ How OSPF Works Here

1. R1 and R2 become OSPF neighbors.
2. They exchange routing information.
3. R1 learns about network 192.168.2.0/24.
4. R2 learns about network 192.168.1.0/24.
5. Packets can now travel automatically between both networks without adding static routes.